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Almost human? A comparative case study on the social media presence of virtual influencers

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ABSTRACT

As virtual agents become prevalent in many domains, virtual influencers have gone live on social media platforms, integrating human networks and interacting with users. Building on research on human-computer interactions, the Uncanny Valley hypothesis, and Computers Are Social Actors paradigm, this paper aims to investigate (1) virtual agents' similarity to humans in terms of behaviour in human networks and (2) reactions to human versus virtual agents in human networks where this interaction is publicly visible. We analyse the posting behaviour of and reactions to one human, one human-like virtual, and one anime-like virtual influencer active on a popular social media platform via text and emoji postings over an 11-month period. We find that, despite the general positive atmosphere of the platform, the human-like virtual influencer receives significantly lower positive reactions, providing evidence for the Uncanny Valley. Additional measures of negative reactions show a similar pattern. We discuss these results within the context of authenticity and social identity on social media, providing recommendations for the implementation of virtual influencers in human social networks.

1. Introduction

Advances in connectivity, portability, and digitalization have led to unprecedented interactivity between humans and virtual agents. Aptly termed the "hyperconnectivity revolution" by Gaines (2019), this new age represents a convergence of technologies that allow for the inclusion of non-human agents into previously human-only spaces (Beer et al., 2015; Gaines, 2019). Autonomous robots are already being integrated into human teams (Demir et al., 2020) and virtual agents equipped with artificial intelligence (AI) are becoming increasingly embedded in fields as diverse as customer service (Krämer et al., 2018), education (Edwards et al., 2019; Potdevin et al., 2020), and healthcare (Beer et al., 2015; Faddoul and Chatterjee, 2020). Virtual conversational agents, such as Siri or Alexa, have been found to convey social benefits via their perceived social presence (Purington et al., 2017), and digital avatars may provide similar benefits on impersonal platforms, enhancing virtual experiences (Hanus and Fox, 2015; Tan and Liew, 2020). There have been accounts of marriage proposals made to Alexa, and men have reportedly married holograms and video game characters (Gersen, 2019). The permeation of virtual agents into human networks is particularly remarkable on social media platforms, as their capacity to provide "the experience of artificial objects as social actors that manifest humanness" is a significant factor in their ability to successfully interact with humans (Thellman et al., 2016).

Nevertheless, to the best of our knowledge, no research has yet looked at virtual agents in human networks where interactions are public. The literature addresses mainly one-on-one interactions between humans and virtual agents, either in private or laboratory settings (Demir et al., 2020; Khan and Sutcliffe, 2014; Krämer et al., 2013, 2018; Li, 2015; Potdevin et al., 2020; Shechtman and Horowitz, 2003; Stein et al., 2020), hence ignoring any potential social identity effects that might arise when the human-agent interaction is visible by others (Purington et al., 2017). Research investigating how people engaging with interactive systems intrigue and attract passers-by, called the honeypot effect (Wouters et al., 2016), might be considered a publicly visible human network context. However, this line of research mostly includes physical public spaces, which attract a relatively limited number of participants compared to online spaces. Given the highly visible and vast nature of online human networks, such as social media, we believe that human-agent interactions in publicly visible online conditions deserve further attention.

One particularly visible type of interaction on social media is that

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with "influencers", i.e. social media users who generate their own content, with the aim of accumulating cultural capital and celebrity in order to shape the opinions or behaviours of their large audiences (Audrezet et al., 2018; Djafarova and Rushworth, 2017). Recently, we have witnessed the emergence of "virtual influencers": agents augmented with digital avatars, designed to look human. These virtual influencers present a particularly interesting context, given that anthropomorphized representations of virtual agents, especially visually attractive ones, elicit social responses and behavioural change in humans (Khan and Sutcliffe, 2014). They are not digital avatars as they are not mere representations of users (Hanus and Fox, 2015), and they are not conversational agents as they are not driven by a user-oriented task (Luger and Sellen, 2016); they are "embodied virtual agents" as defined by Tan and Liew (2020). These virtual influencers are presented similarly to human influencers, with their own public personas and story lines, which allow for greater interaction between users and influencers in the virtual environment (Hanus and Fox, 2015). In some cases – such as Shudu, a computer-generated model – the creators make no effort to hide the digital origins of the influencer, while in other cases – such as LilMiquela – the true nature of the influencer is at best ambiguous, at worst misrepresented (Klein, 2020).

As the line between human and virtual agents becomes increasingly blurred, we aim to explore two research questions:

RQ1: How similar is the behavior of virtual agents to humans in publicly visible human networks?
RQ2: How do humans react to virtual agents embedded in publicly visible human networks?

To answer the former, we compare the posting behaviour of three accounts on social media: one human influencer, one human-like virtual influencer, and one anime-like virtual influencer; to answer the latter, we consider the content of social media users' comments vis-à-vis these three influencers.

In order to understand this particular instance of human-virtual agent interaction, we first introduce the concept of influencers in the following section. We then proceed to review existing literature on virtual agents, as well as conversational agents and digital avatars, as virtual influencers can be understood as an amalgam of those often interchangeable concepts (Hanus and Fox, 2015; Luger and Sellen, 2016). As our review points to both the potential acceptance and rejection of virtual agents in human networks, we then present an exploratory online case study, comparing the posting behaviours of and reactions to the human and virtual influencers.

2. Related work

2.1. From human to virtual influencers

In addition to fulfilling social, informational, and entertainment needs (Casaló et al., 2018; Xu et al., 2017), social media provides the unique opportunity to interact with celebrities and brands (Djafarova and Rushworth, 2017; Hanus and Fox, 2015; Hwang and Zhang, 2018; Kim and Drumwright, 2016). These interactions have spawned their own type of celebrity, famous primarily for their social media presence rather than achievements in offline public life or performing arts (Djafarova and Rushworth, 2017; Hwang and Zhang, 2018). Among these, a number of opinion leaders have emerged, called *influencers:* specialists in personal branding, who cultivate a unique public image visible via their online presence (Khamis et al., 2017). These microcelebrities have created an influencer marketing industry that is estimated to reach $15 billion by 2022 (Rahal, 2020).

Social media enables bidirectional communication where receivers can also participate in the activities posted by the senders in a bottom-up fashion, as opposed to traditional media where communication is diffused top-down (Dijkmans et al., 2015; Hanus and Fox, 2015). Influencers take advantage of this mass interpersonal communication to make connections with their audience and be seen as more relatable than mainstream celebrities (Djafarova and Rushworth, 2017). Influencers have become particularly attractive to marketing professionals, as they provide a less invasive advertising medium (Johnson et al., 2019). Naturally, the influencer must be seen as credible, attractive, trustworthy, and knowledgeable in the area of the endorsed product to be convincing (Djafarova and Rushworth, 2017). Additionally, it is important that the endorsement by the influencer be authentic, in line with both the company/product and the influencer's public image (Audrezet et al., 2018).

However, some scholars argue that there is an inherent ethical problem with the idea of "branding" an individual, as well as potential problems of image consistency amid the omnipresent risk of the occasional indiscretion (Khamis et al., 2017). Influencers are also gaining negative reputations for excessive demands that may turn off brands or certain user demographics (McIntosh, 2019). Additionally, many question whether the authenticity inherent in influencers has been tarnished, with suspicions of staged photos and automated content becoming more prevalent (Kleinman, 2019; Russon, 2018).

Virtual influencers remove many of these concerns: as they are not humans, there is no ethical issue of branding attached; the image can remain consistent, and the risk of indiscretions is minimized as they don't exist offline, so their "behaviour" and image can be calibrated in the background (Tan, 2019). The virtual influencer maintains their influential status and persuasive effectiveness via the attractiveness stereotype, human-like functionality, and audio-visual features (Faddoul and Chatterjee, 2020; Khan and Sutcliffe, 2014), while potentially even mitigating concerns about authenticity: as a virtual influencer is "authentically fake" (Wills, 2019), the user is well aware that they are consuming staged content.

2.2. User engagement

Virtual influencers provide the potential for a new realm of user engagement, defined as a "mental state (…) that regards his/her connection with the other participant and the interaction itself" (Glas and Pelachaud, 2018). Engagement with human agents via digital media has been found to fulfil the needs for personal identity, diversion, social relatedness, and autonomy (Gaines, 2019; Hanus and Fox, 2015; Partala, 2011; Sheldon and Bryant, 2016); we argue that virtual influencers, much like human influencers, can similarly fulfil these needs on social media.

Firstly, interactions with influencers can aid in developing and showcasing one's personal identity; this is especially true in highly visible contexts such as social media. Social identity theory argues that one's perceived membership in social groups constitutes a part of one's identity and allows for the identification of clear in/out groups (Edwards et al., 2019; Tajfel, 1978). Users are often aware that their posting behaviour on social media platforms will be visible to others, particularly on sites like Instagram where the majority of users do not opt for private accounts and choose to leave their content as publicly visible (Waterloo et al., 2018). Therefore, the groups or other users with whom individuals choose to interact and in what fashion signals the social and relational identities, respectively, that they wish to cultivate (Edwards et al., 2019; Pan et al., 2017). Influencers in general are associated with fashion and lifestyles domains (Audrezet et al., 2018), but virtual influencers may provide additional signalling value with respect to their follower's innovativeness, open-mindedness, or being in line with the latest trends. The social identity that comes from signalling any of these attributes may drive exploratory behaviour on social media, leading users to seek out virtual influencers (Pan et al., 2017).

Secondly, virtual influencers fulfil the need for diversion, as they offer a view into a different world. Similar to the investment that viewers make in a favourite movie or television character, the influencer's life and various "storylines" may serve as an escape from the user's

seemingly mundane life. Although such diversion may also be possible from human influencers, virtual influencers bridge real and imaginary worlds, which may provide additional entertainment or escape benefits. Banks and Cole (2016) report that military members used video games to escape from daily stresses, but, interestingly, preferred military-themed games, implying that, ideally, fictional worlds should be based on reality (Dill-Shackleford et al., 2016). Comparable to conversational agents that provide playful experiences to its users (Luger and Sellen, 2016), virtual influencers allow the user to immerse themselves in an alternate reality, albeit one sufficiently similar to, and still based in, the real world.

This diversion and immersive experience may serve as a social interaction, regardless of the fact that the interaction is with a virtual agent (Gratch et al., 2007; Potdevin et al., 2020). Furthermore, social media allows for a more interactive type of communication where the audience reaction can alter the poster's behaviour (Bareket-Bojmel et al., 2016). Therefore, users may feel that they are developing a more intimate relationship with their preferred influencer(s), where not only is the user subject to communication from the influencer, but they are able to communicate back to the influencer as well. At the extreme, this may result in the perception of a parasocial relationship where the user feels a mutual awareness or friendship with the influencer and may even position themself (the user) as part of the influencer's social world (Dibble et al., 2016; Hwang and Zhang, 2018). The phenomenon is well documented with celebrities and human influencers (Hwang and Zhang, 2018) and there are reasons to believe that virtual influencers may elicit a similar reaction. Gratch et al. (2007) showed that virtual agents are able to create "rapport", a feeling of connectedness via socio-emotional processes. Research by Krämer et al. (2013, 2018) has found that interactions with virtual agents can meet some of the same social interaction needs as human-human interactions, and that humans even engage in mimicry with virtual agents, similarly as with human counterparts. Furthermore, Felnhofer et al. (2019) found that virtual agents were rated similarly on social presence as avatars controlled by humans, concluding that the mere presence of these virtual agent acted as a social support, mitigating subsequent perceived stress and reductions in self-esteem resulting from a standardized stress test.

Finally, interactions on social media with virtual influencers may fulfil needs of autonomy, which are required for the user to experience an intrinsic enjoyment of the interaction (Downie et al., 2008; Ryan and Deci, 2000). Much as with human influencers, the individual is choosing to interact with a persuasive agent and consume any promotional or advertising content that may appear (Hanus and Fox, 2015). Additionally, by interacting with a known virtual agent, the user is choosing to be drawn into a virtual world, with no risk of being fooled by inauthentic storylines or staged photographs.

2.3. User reaction to virtual influencers

The above discussion illustrates that there are reasons to expect that virtual agents can fulfil many of the same needs as humans in human networks. However, arguments exist on both sides for the specific nature of the reaction of humans to virtual agents in such contexts.

On the one hand, there is a long history of research showing that human beings tend to anthropomorphize machines and abide by social rules in their interactions with them (Feine et al., 2019; Krämer et al., 2013; Nass and Moon, 2000; Purington et al., 2017; Schroeder and Epley, 2016). The Computers Are Social Actors (CASA) paradigm suggest that humans apply social heuristics of human interaction to their interactions with computers via the mindless attribution of social traits (Edwards et al., 2019; Nass et al., 1995; Nass and Moon, 2000). By looking at norms of categorization, politeness, reciprocity, self-disclosure, and the effects of premature labelling and homophily, research has consistently found that people react to computers in much the same way that they would be expected to react to humans (Fogg and Nass, 1997; Nass et al., 1995; Nass and Moon, 2000). Furthermore, the anthropomorphising of virtual agents can aid in decreasing the uncertainty of interactions and increasing their perceived social presence (Schroeder and Epley, 2016), thereby rendering the human-virtual agent interaction more similar to human-on-human ones (Edwards et al., 2019).

Indeed, more recent studies have found that people react socially, emotionally, cognitively, and behaviourally to virtual agents similarly as they do to other humans, and even show brain activation in regions related to emotional and interpersonal experiences when interacting with virtual agents (Krämer et al., 2015; Von der Pütten et al., 2010). Potdevin et al. (2020) found that people projected themselves into anthropomorphic interactions with machines through the feeling of social presence, while Kwon et al. (2011) established that older users sought out digital agents' presence to alleviate their distrust in online shopping, implying that virtual agents may be accepted and treated as a peer in human networks, even by the least likely of audiences (Venkatesh and Bala, 2008). Li (2015) found that, when users interacted with a robot on a screen, there was no difference between physical versus virtual embodiment. Although the robots in the studies reviewed in Li (2015) were not highly anthropomorphized, the results nevertheless point to the conclusion that physical embodiment is not necessary to elicit favourable psychological responses. Finally, Edwards et al. (2019) combined CASA with social identity theory to demonstrate that age group stereotypes (e.g. older is wiser) were applied to a virtual agent based on vocal stimuli.

On the other hand, Shechtman and Horowitz (2003), drawing on interpersonal theory, found that people reacted differently depending on whether they believed they were interacting with a computer or a human, putting more effort into their conversation when they believed they were speaking with a human partner. There is also evidence that especially human-like robots and avatars, particularly when coupled with a perception of human-like cognitive abilities, can engender negative affective reactions (MacDorman, 2019; Stein et al., 2020). This idea is captured in the Uncanny Valley (UV) hypothesis (Mori, 1970; Mori et al., 2012), which argues that as non-human entities such as avatars or robots become more human-like, they may reach a point where they elicit a negative reaction from users, due to feelings of an uncanny resemblance to humans (Katsyri et al., 2017; Mathur et al., 2020; Wiese and Weis, 2020). When an agent is almost human looking, the affinity with it decreases (Beer et al., 2015) as the imperfections in this resemblance make people uncomfortable (Li, 2015). Additionally, social acceptance of virtual agents is hindered by the cultural demonization of advanced technologies, especially in Western fiction (Stein et al., 2020), potentially making people more sceptical in their interaction with virtual agents. Finally, attempts at mechanistic dehumanization may serve to strip away human characteristics, leading to a distancing from and indifference to the non-human agent (Haslam, 2006).

Given these contradictory arguments, we do not propose any specific hypotheses. Instead, we proceed with an exploratory study to answer our two main research questions by analysing influencers on Instagram. First, we analyse how human vs. virtual influencers differ in terms of posting behaviour. We then investigate user reactions by analysing how comments differ with respect to human vs. virtual influencers.

3. Methodology

Instagram is our focal social media platform because it is one of the fastest growing social network sites (Pineda et al., 2020; Sheldon and Bryant, 2016) and the most used platform by influencers (Casaló et al., 2018). Instagram consists of visual content, in the form of photos or videos, often augmented by short text. It is a highly visual platform (Casaló et al., 2020; Johnson et al., 2019), based largely around one's personal identity (Sheldon and Bryant, 2016) and where the majority of users choose to maintain the information they post as public (Waterloo et al., 2018). Additionally, followers committed to an online community

are more prone to interact on that particular platform (Casaló et al., 2020). Hence, given that influencers came to prominence primarily via Instagram and attractive visual representation of the virtual agent is an important factor in human-agent interaction (Khan and Sutcliffe, 2014), this platform is particularly well suited to an analysis of user reaction to them.

We collected both influencer posts and a sample of all public comments to these posts, from followers and non-followers alike, to avoid any potential bias arising from parasocial relationships or an increased familiarity with the influencer, or from a tendency to comment positively stemming from following the influencer. Even though there are multiple languages identified in the comments, the language of analysis is English due to both the dominance of English in the interaction with the influencers and the limitations of the analysis tools in processing non-English languages.

We also take note of the controversy around the usage of publicly available online data (Moreno et al., 2013). The data from all of the influencers we considered fall under the category of "public figure" as all cases of influencers are verified accounts with "Blue Check Mark", a label that verifies their public status. Additionally, user data is used anonymously, and any handles attached to comments cannot be used to identify specific individuals. Thus, we are confident that these measures remove any potential data protection concerns.

The UV hypothesis argues for the existence of a "dip" in the perceiver's affective reaction to the entity as a function of their human likeness (Mori, 1970; Mori et al., 2012). In order to be able to identify this descent, we identified three influencers to study: one who was clearly not human and represented what Mori (1970) termed the "Toy Robot" end of the spectrum; one influencer who appeared human-enough to give rise to suspicion about her virtual vs. human nature; and one human influencer (HI).

3.1. Anime-like virtual influencer

We identified two clearly non-human virtual influencers active on Instagram: B. the Bee influencer[1] and Noonoouri[2]. As B. is not a humanoid entity, we chose Noonoouri as the best representative of a clearly non-human entity that has been nonetheless anthropomorphised to fit into a human social network. At the time of this writing, Noonoouri had approximately 371,000 followers and posted visibly computer-generated photos reflecting an artistic, urban style.

3.2. Human-like virtual influencer

Virtual agents on social media such as bots, targeted ads, etc. are not a new phenomenon. However, with the advancement of digital technologies, computer generated images (CGI) are more and more human-like and virtual agents are increasingly capable of simulating human content. Making use of these technological advancements, a Los Angeles-based digital company, brud, introduced digital personas on social media platforms in 2017 with three "products": LilMiquela[3], Blawko22[4], and Bermudaisbea[5] (Shieber, 2018). Described as virtual influencers by the media, these personas rose to prominence after an internally orchestrated feud in April 2018, where LilMiquela's account was "hacked" by Bermudaisbae, with the latter exposing the former as a CGI bot (Sims, 2018). By 2019, human-like virtual influencers (HVI) were gaining traction, with the appearance of additional accounts. In order to select our focal HVI, we identified five virtual influencers who portrayed a similar style to Noonoouri and each other: Immagram[6], LilMiquela, Liam Nikuro[7], Bermudaisbae, and Blawko. We asked 402 Instagram users, recruited via the Prolific online research participant pool (45.5% female, average age 23.3 years old, SD = 4.00), to rate these influencers, plus Noonoouri, on their human likeness. For each of the six virtual influencers, participants were asked to rate, on a scale of 1 = Not at All to 7 = Extremely, their response to the question "Please rate the extent to which you feel each of the following terms describes the influencer depicted in the picture above", followed by a list of adjectives[8]. For the construct of "humaneness", we averaged the scores for each influencer on the adjectives of "genuinely human", "cartoonish", and "exaggerated" (latter two variables reverse coded; Katsyri et al., 2017). As expected, Noonoouri scored the lowest, thereby validating our choice of the clearly non-human data point. Given her similarity to anime characters (presenting human-adjacent characteristics such as exaggerated eyes, small mouth, etc.; Yang et al., 2020), we will refer to her as the "anime-like virtual influencer" (AVI) to distinguish her from the HVI.

In order to choose the virtual influencer who was as human-like as possible, yet ambiguous enough to elicit doubt amongst users, we looked at the average score on this humaneness measure across all six profiles. LilMiquela had the humaneness score the closest to this average, and was therefore designated as our HVI, representing the mid-point between Noonoouri and the HI.

With over three million followers on Instagram, her primary platform, LilMiquela has received press coverage in the New York Times (Hsu, 2019), Washington Post (Holley, 2018), Business Insider (Hoffower, 2019), and Vox (Tiffany, 2019). She is presented as an urban, Latin-American young adult female, who engages in human activities such as travel, artistic creation (music), spending time with others (humans and other digital entities), and shopping. She has collaborated with major brands such as Chanel, Coach, Balenciaga, and Prada.

3.3. Human influencer

Using LilMiquela as our virtual referent, we identified the following criteria in our search for an HI with similar key traits and therefore expected to attract a similar audience:

- Woman aged 18–29
- Instagram-famous
- Category: Lifestyle content creator
- Residing in US West Coast
- Financially independent
- No children
- Heterosexual
- Engagement in artistic creation

We conducted an online search as well as tapped into our personal network to identify such influencers. We identified six social media influencers matching the majority of our criteria: Marta Cygan[9], Rachel Levin[10], Juice Gee[11], Jenn McAllister[12], Megan DeAnglelis[13], and Megan

[1] https://www.instagram.com/bee_nfluencer/
[2] https://www.instagram.com/noonoouri/
[3] https://www.instagram.com/lilmiquela/
[4] https://www.instagram.com/blawko22/
[5] https://www.instagram.com/bermudaisbae/
[6] https://www.instagram.com/imma.gram/
[7] https://www.instagram.com/liam_nikuro/
[8] This pilot study was used to measure opinions of virtual influencers on several dimensions for a different project, therefore all of the pilot study measures are not described here but available from the first author upon request.
[9] https://www.instagram.com/lifeofboheme/
[10] https://www.instagram.com/rclbeauty101/
[11] https://www.instagram.com/juicegee/
[12] https://www.instagram.com/jennxpenn/
[13] https://www.instagram.com/megandeangelis/

Rienks[14].

In order to choose amongst these HIs, we recruited an additional 151 Instagram users, using the same portal as for the first pilot study (44.5% female, average age 23.54, SD = 4.23). Each participant was presented with six pairs of photos (one of LilMiquela and one of each human influencer, all taken from the influencers' Instagram account) and asked to rate the extent to which they agreed (1 = Strongly Disagree, 7 = Strongly Agree) with the three following statements: "These two influencers have a similar personal style", "These two influencers post similar content", and "These two influencers have similar followers". Given that Marta Cygan was rated as closest to LilMiquela in two of these three categories and rated second in the third category (similarity of content posted), she was therefore chosen as case HI, representing the final data point in our study. At the time of this writing, Marta Cygan had approximately 239,000 followers and posted photos primarily depicting a lifestyle centred on fashion and beauty in an urban setting.

These three influencers are shown in Fig. 1. Going forward, we will refer to Noonoouri, as AVI (anime-like virtual influencer); LilMiquela as HVI (human-like virtual influencer); and Marta Cygan as HI (human influencer).

3.4. Data collection

3.4.1. Time period

To minimize any confounding effects, we choose to limit our study to posts and comments made during a specific time period. In order to be far enough away from the disruptive event of LilMiquela's "reveal" as a virtual agent (described in Section 3.1, dated April 2018), we set the start date of our data collection as January 1st, 2019. In order to avoid any effects of the American holiday season (Hu, 2013), we decided to stop our data collection two days before Thanksgiving, on November 26, 2019.

3.4.2. URL extraction

The observational unit for this study is the text from Instagram posts made by AVI, HVI, and HI, as well as a sampling of public comments made in response to these posts. Once the Instagram accounts were identified, we proceeded to extract links for each post made by AVI, HVI, and HI. For this purpose, we used a commercially available online tool, Instagram Posts Extractor. This allowed us to collect all of the URLs of each post by AVI, HVI, and HI, the content of the post, number of comments, likes and video views (for posts containing videos), date and location stamps. The final data set contains 381 posts for AVI, 281 posts for HVI, and 349 posts for HI falling within our target time period. These posts received 72,709, 140,564, and 17,741 total comments, respectively.

3.4.3. Comment extraction

We made use of open data material to develop a Python code that extracts comments from each unique post URL. The comment extraction process was conducted as a function of each collected url. First an url was launched, then the comments were extracted by loading them incrementally. However, due to Instagram Application Programming Interface (API) restrictions (Constine, 2018), we were not able to collect all comments for every post. The extraction for each post proceeded until the page no longer validated the extraction and therefore terminated access to the url, either because of time restrictions, or limits on the number of times a given account can access a url. Hence, our final sample of comments represents a random sampling of the comments each influencer received in response to her original post.

Instagram proprietary algorithms prioritize the order of comments displayed based on highest engagement (liking), commenting user's profile status, or the account user's past activity (Read, 2018). We used an account specifically set up for extraction purposes, without prior activity. Therefore, we do not expect this latter factor to have had an impact. We recognize that we may have an overrepresentation of highly liked comments in our sample, however these possibly represent comments that are most relevant to the initial post. Additionally, given that these sampling restrictions would have applied equally to all three influencers, we do not expect any systematic bias to have presented in the final sample of comments.

This extraction process resulted in a final sample of 12,889 comments for AVI, 30,536 comments for HVI, and 5402 comments for HI. Given the total number of comments received by each influencer, these numbers represent a sample of 18%, 22% and 30% of comments, respectively.

3.5. Data preparation

Data was comprised of both text and emoji - typographic facial representations used in digital communication (Cramer et al., 2016; Fernández-Gavilanes et al., 2018). Emoji use is prevalent on digital platforms and provides a complementary context to the text, however the intended messaging is not always straightforward (Cramer et al., 2016; Tigwell and Flatla, 2016). Hence, to analyse the sentiments conveyed by emoji use, we referred to the "Emoji Lexica" created by Fernández-Gavilanes et al. (2018), who categorize both facial and non-facial emoji as positive, negative, or neutral, based on the valence assigned to the post in which each emoji was used. Some emoji showed the same score for multiple valence categories and were analysed as such. Where applicable, we added variations of emoji missing from the original lexica (e.g. different skin tones).

Based on this specific context and our theoretical framework, we created several custom lists of emoji or terms that were used in the analysis, to address the platform-specific differentiation of emoji use (Tigwell and Flatla, 2016). In each case, emoji in a custom list where removed from the positive/negative/neutral lists if necessary:

- Doubt: We wanted to capture terms that articulated the non-human nature of the virtual influencers (robot, avatar, creepy, fake, etc.).
- Encouragement: To capture any potential CASA effects, we chose terms and emoji related to encouragement and supports, such as "queen" and "congratulations", along with emoji representing crowns, royalty, bees and bombs (within the context of "queen bee" and "you're the bomb").
- Heart emoji: We separated out all emoji that represented hearts or love, with the exception of the heart-eyes emoji (see below) and the broken heart emoji (which remained in the negative emoji list) to create an independent measure of positive affective reactions.
- Heart-eyes emoji: This emoji warranted special attention given its conveyance of "enthusiastic feelings of love, infatuation, and adoration" (Emojipedia Pty Ltd., 2019). These sentiments are much stronger than the positive emotions conveyed by other facial emoji, but distinct from the hearts emoji in our custom category.

We then proceeded to code each post and user comment for the presence/absence of items falling into each of these categories, plus the three emoji valence lists (1 = content of category present at least once, 0 = otherwise). We also created a composite variable to capture the general existence of an emoji in a post or comment, which was equal to 1 if the dummy variable associated with any of the emoji-only categories was 1, 0 if all were 0, resulting in a total of 8 dummy variables.

For linguistic analysis, we relied on the categories available in LIWC 2015 (Pennebaker et al., 2015). Specifically, we were interested in all of the affective categories, which include overall tone, positive, and negative emotions; this latter category includes three discrete emotions: anxiety, anger, and sadness. We were also interested in the categories representing affiliation and different types of relationships (affiliation, family, friendship), gender specific references (male, female) and

[14] https://www.instagram.com/meghanrienks/

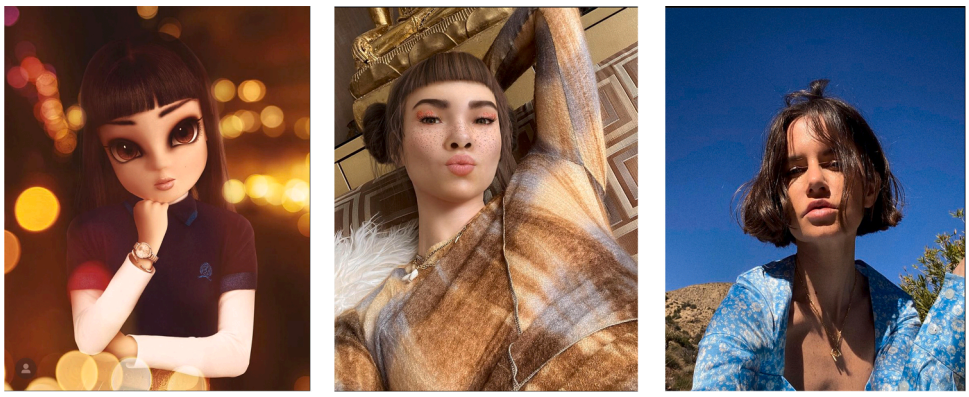


Fig. 1. Noonoouri as AVI, LilMiquela as HVI, Marta Cygan as HI.

informal speech (swearing, netspeak, nonfluid speech, filler words). More detailed explanations of the categories, along with example words for each, are available in Pennebaker et al. (2015).

4. Results

We conducted two separate analyses, looking first at the posting behaviour of our three influencers, before turning our attention to the user comments to these influencers. We conducted ANOVA analyses to determine whether the posting behaviour of and the reactions to the influencers differed, comparing each influencer with one other. This resulted in three dyad comparisons for each variable of interest (HVI vs. HI, HVI vs. AVI and HI vs. AVI). We then conducted Tukey's test to evaluate the significance of differences between each dyad. However, given the large sample sizes, the results of these analyses may give artificially small *p*-values, resulting in even very small differences being identified as significant. Therefore, in addition to *p*-values, we report absolute values of Cohen's *d* as a measure of effect size, relying on the traditional values of $d = 0.20$, 0.50, and 0.80 as the thresholds for small, medium, and large effect sizes, indicated in the manuscript as *, **, ***, respectively (Cohen, 1988). We used an equation to calculate Cohen's *d* that took into account the differences in subgroup sample sizes.

4.1. Comparison of influencer posts

AVI has the most posts (381) during our defined time period. HI was in second place with 349 posts, and HVI had the fewest at 281 posts. AVI and HVI showed similar usage of videos (37 vs. 36, respectively), which is considerably higher than that of HI, who posted 13 videos during the same period.

Table 1 shows the results of mean comparisons regarding common social media engagement metrics (Gräve, 2019; Jang et al., 2015). HVI receives the most likes per post and views per video, significantly more than the others (Likes: $d_{HVI, HI} = 1.47^{***}$, $d_{HVI, AVI} = 0.24^{*}$, $d_{HI, AVI} = 2.02^{***}$; Video views: $d_{HVI, HI} = 1.43^{***}$, $d_{HVI, AVI} = 0.63^{**}$, $d_{HI, AVI} = 1.09^{***}$). In both cases, AVI has the second highest number, and HI the lowest. This trend does not hold for comments per post however, as in this case AVI has the highest number of comments, followed by HVI and HI ($d_{HVI, HI} = 0.32^{*}$, $d_{HVI, AVI} = 0.73^{**}$, $d_{HI, AVI} = 1.35^{***}$). Therefore, it appears that the virtual influencers overall receive the most engagement in terms of social media engagement metrics.

Table 2 shows the results of mean comparisons of the posting behaviour of the three influencers in terms of emoji. In general, HVI uses significantly more of all three of the emoji types, with HI using the least positive and neutral emoji (Positive emoji: $d_{HVI, HI} = 1.35^{***}$, $d_{HVI, AVI} = 0.78^{**}$, $d_{HI, AVI} = 0.44^{*}$; Neutral emoji: $d_{HVI, HI} = 1.33^{***}$, $d_{HVI, AVI} = 1.14^{***}$, $d_{HI, AVI} = 0.23^{*}$). For the negative emoji, both HI and AVI have an average per post of zero[15], while HVI has an average of 0.12 negative emoji per post ($d_{HVI, HI} = 0.56^{**}$, $d_{HVI, AVI} = 0.57^{**}$, $d_{HI, AVI} =$ N/A).

In terms of the linguistic content as displayed in Table 3, HI clearly posts significantly shorter posts than the two virtual influencers ($d_{HVI, HI} = 1.50^{***}$, $d_{HVI, AVI} = 0.16$, $d_{HI, AVI} = 1.45^{***}$). Similarly, the posts by both virtual influencers are more positive than those of HI, as evidence by scores in both the overall tone ($d_{HVI, HI} = 1.14^{***}$, $d_{HVI, AVI} = 0.17$, $d_{HI, AVI} = 1.37^{***}$) and the positive emotion categories ($d_{HVI, HI} = 0.67^{**}$, $d_{HVI, AVI} = 0.05$, $d_{HI, AVI} = 0.69^{**}$).

Interestingly, HVI's posts show significantly more negative emotion terminology than either AVI or HI ($d_{HVI, HI} = 0.33^{*}$, $d_{HVI, AVI} = 0.37^{*}$, $d_{HI, AVI} = 0.07$). In terms of the three discrete emotion categories available in LIWC, the three influencers show similar patterns for anxiety, anger, and sadness, in that HVI expresses each of these emotions the most (Anxiety: $d_{HVI, HI} = 0.35^{*}$, $d_{HVI, AVI} = 0.22^{*}$, $d_{HI, AVI} = 0.20^{*}$; Anger: $d_{HVI, HI} = 0.27^{*}$, $d_{HVI, AVI} = 0.16$, $d_{HI, AVI} = 0.24^{*}$; Sadness: $d_{HVI, HI} = 0.01$, $d_{HVI, AVI} = 0.14$, $d_{HI, AVI} = 0.05$), although these differences were not significantly different between the three influencers for sadness. Our results also show that HVI uses significantly more terminology than at least one of the other influencers when it comes to categories of gender (Male: $d_{HVI, HI} = 0.21^{*}$, $d_{HVI, AVI} = 0.29^{*}$, $d_{HI, AVI} = 0.01$; Female: $d_{HVI, HI} = 0.34^{*}$, $d_{HVI, AVI} = 0.14$, $d_{HI, AVI} = 0.23^{*}$), friends ($d_{HVI, HI} = 0.29^{*}$, $d_{HVI, AVI} = 0.13$, $d_{HI, AVI} = 0.17$), family ($d_{HVI, HI} = 0.29^{*}$, $d_{HVI, AVI} = 0.14$, $d_{HI, AVI} = 0.23^{*}$), and affiliation in general ($d_{HVI, HI} = 0.30^{*}$, $d_{HVI, AVI} = 0.31^{*}$, $d_{HI, AVI} = 0.13$).

Similarly, in terms of the language type, HVI uses significantly more swear words than AVI ($d_{HVI, HI} = 0.02$, $d_{HVI, AVI} = 0.25^{*}$, $d_{HI, AVI} = 0.08$). HVI also uses significantly more netspeak (e.g. lol, btw) and significantly more filler speech (e.g., I mean, you know) than both AVI and HI (Netspeak: $d_{HVI, HI} = 0.54^{**}$, $d_{HVI, AVI} = 0.52^{**}$, $d_{HI, AVI} = 0.11$; Filler: $d_{HVI, HI} = 0.23^{*}$, $d_{HVI, AVI} = 0.23^{*}$, $d_{HI, AVI} =$ N/A). There were no significant differences between any of the three dyads for nonfluid speech (e.g. er, umm).

4.2. Comparison of user comments

Our analysis of posting behavior allows us to understand the stimuli to which users are reacting with their comments. Despite her using the

[15] Although it is possible that some posts by these influencers used negative emoji, the proportion of them within the overall data set was too small to register within the first two decimal places. We therefore treat these values as zero.

Table 1
Engagement Metrics.

	AVI 381 (37 videos)		HVI 281 (36 videos)		HI 349 (13 videos)	
	M	SD	*M*	SD	*M*	SD
Avg comments per post (scaled)	507.46	275.95	277.90	356.90	189.98	182.26
Average likes per post (scaled)	27,034.88	8,702.99	30,376.11	18,406.19	10,440.67	7,613.92
Average views per video (scaled)	213,427.26	142,410.56	321,364.68	196,129.86	75,725.60	5,9413.00
Use of video	0.10	0.30	0.13	0.33	0.04	0.19
	HVI vs. AVI		AVI vs. HI		HVI vs. HI	
	p-value	Cohen's d	*p*-value	Cohen's d	*p*-value	Cohen's d
Avg comments per post (scaled)	0.00	0.73	0.00	1.35	0.00	0.32
Average likes per post (scaled)	0.00	0.24	0.00	2.02	0.00	1.47
Average views per video (scaled)	0.01	0.63	0.02	1.09	0.00	1.43
Use of video	0.33	0.10	0.01	0.24	0.00	0.34

Table 2
Emoji analysis of influencer posts.

	AVI		HVI		HI	
	M	SD	*M*	SD	*M*	SD
Positive Emoji	0.35	0.48	0.72	0.45	0.17	0.37
Negative Emoji	0.00	0.00	0.12	0.33	0.00	0.00
Neutral Emoji	0.05	0.21	0.46	0.50	0.01	0.09

	HVI vs. AVI		HI vs. AVI		HVI vs. HI	
	p-value	Cohen's d	*p*-value	Cohen's d	*p*-value	Cohen's d
Positive Emoji	0.00	0.78	0.00	0.44	0.00	1.35
Negative Emoji	0.00	0.58	1.00	NA	0.00	0.56
Neutral Emoji	0.00	1.14	0.19	0.23	0.00	1.33

most positive emoji and tone, we see that not only did HVI receive the fewest positive emoji in comments to her posts, but that these effect sizes are medium sized when comparing her to the other two influencers, as seen in Fig. 2. More importantly, the effect size comparing AVI and HI is not significant ($d_{HVI, HI} = 0.51^{**}$, $d_{HVI, AVI} = 0.65^{**}$, $d_{HI, AVI} = 0.13$). We see a similar pattern for negative and neutral emoji usage, although for the former there is also a small difference between the negative emoji in comments for AVI and HI ($d_{HVI, HI} = 0.18$, $d_{HVI, AVI} = 0.41^{*}$, $d_{HI, AVI} = 0.21^{*}$), while the effect sizes for the latter are stronger for HVI and HI as well as HVI and AVI ($d_{HVI, HI} = 0.30^{*}$, $d_{HVI, AVI} = 0.49^{*}$, $d_{HI, AVI} = 0.18$).

Our analysis also included two additional emoji that signal positive affect: hearts and heart-eyes. In both cases, as shown in Fig. 3, we see evidence supporting the UV hypothesis (Hearts: $d_{HVI, HI} = 0.37^{*}$, $d_{HVI, AVI} = 0.56^{**}$, $d_{HI, AVI} = 0.18$; Heart-eyes: $d_{HVI, HI} = 0.48^{*}$, $d_{HVI, AVI} = 0.36^{*}$, $d_{HI, AVI} = 0.07$). Additionally, the user reactions in terms of using doubting terminology clearly show that HVI presents the most ambiguous appearance, with some doubt surrounding AVI, and no doubt regarding HI ($d_{HVI, HI} = 0.34^{*}$, $d_{HVI, AVI} = 0.32^{*}$, $d_{HI, AVI} = 0.13$).

In terms of linguistic analysis (Table 5), we see that HVI received the longest comments, on average, followed by AVI and then HI, with these effect sizes being small to medium in size ($d_{HVI, HI} = 0.53^{**}$, $d_{HVI, AVI} = 0.35^{*}$, $d_{HI, AVI} = 0.28^{*}$). Although there were no significant differences in terms of overall tone, the positive and negative emotion categories provide further evidence for the UV hypothesis, as shown in Fig. 4. Users reacted with approximately equal amounts of positive affect terms towards AVI and HI, but with significantly lower use of such terms with respect to HVI ($d_{HVI, HI} = 0.47^{*}$, $d_{HVI, AVI} = 0.36^{*}$, $d_{HI, AVI} = 0.07$). For negative affect related terms, the graph shows an inverse UV effect with a visible increase showing that HVI received the most negative affect terminology. Although these effect sizes do not reach the threshold of 0.20 ($d_{HVI, HI} = 0.15$, $d_{HVI, AVI} = 0.16$, $d_{HI, AVI} = 0.01$), it is worth nothing that the differences between HVI and the other two influencers are very close to this threshold, while the difference between AVI and HI is 0.01, with a p-value clearly signaling insignificance of the difference in effect sizes (despite the inflated t-test statistics resulting from the large sample sizes involved).

We see a similar pattern for the anger category, which shows a much higher anger response to HVI's comments (Anger: $d_{HVI, HI} = 0.15$, $d_{HVI, AVI} = 0.15$, $d_{HI, AVI} = 0.02$) with the *p*-value for the mean difference between HI and AVI again indicating clear insignificance of the effect size differences (Fig. 5). For the other two discrete negative emotion categories of anxiety and sadness, the effect sizes were below 0.10, although the pattern for anxiety warrants further examination in the discussion section.

Another potential indicator of a negative emotional reaction could be the category of swear words. As Fig. 6 shows, we once again see the highest value for HVI ($d_{HVI, HI} = 0.19$, $d_{HVI, AVI} = 0.15$, $d_{HI, AVI} = 0.09$), with an effect size approaching the 0.20 threshold for the comparison of HVI and HI.

In general, the overall informal category, and the netspeak category in particularly, showed a similar pattern to the swear category, albeit with significant effect size in terms of the differences between HVI and HI (Informal: $d_{HVI, HI} = 0.23^{*}$, $d_{HVI, AVI} = 0.18$, $d_{HI, AVI} = 0.07$; Netspeak: $d_{HVI, HI} = 0.25^{*}$, $d_{HVI, AVI} = 0.19$, $d_{HI, AVI} = 0.08$). However, these results should be interpreted with caution as this may be a case of users mimicking the influencers own linguistic style given that the influencer posts showed a very similar pattern. The final two categories of informal language, non-fluid speech and filler words, did not show any effect sizes approaching the predetermined thresholds for significance of the Cohen's *d* statistic.

In terms of gendered language, the female category, shown in Fig. 7, showed an effect size approaching the lower threshold for comments in reaction to HVI vs. HI ($d_{HVI, HI} = 0.19$, $d_{HVI, AVI} = 0.09$, $d_{HI, AVI} = 0.12$), but not when comparing HVI and AVI, nor when comparing HI and AVI. Although HVI also received the most male related terminology in her comments, the differences amongst the three influencers did not approach the small-effect size threshold of 0.20.

Finally, in order to identify the CASA effect within user comments, we looked for a lack of significant differences in the use of terminology related to the "encouragement" category we had created and the family, friends, and affiliation LIWC categories. There were in fact no significant differences between dyads on encouragement (Table 4, Fig. 3) nor the friends or family relational categories (Table 5, Fig. 7). The affiliation category showed the similar uncanny valley dip, with a slightly significant difference between HVI and HI ($d_{HVI, HI} = 0.20^{*}$, $d_{HVI, AVI} = 0.06$, $d_{HI, AVI} = 0.13$)

5. Discussion

This work contributes to the UV literature by building on past research comparing peoples' interactions with human vs. virtual agents

Table 3
Linguistic analysis of influencer posts.

	AVI		HVI		HI	
	M	SD	*M*	SD	*M*	SD
Word Count	38.24	32.38	33.23	28.58	4.20	3.67
Tone	65.71	33.20	59.97	35.72	29.40	16.31
Positive Emotion	3.98	4.43	4.23	5.52	0.87	4.57
Negative Emotion	0.25	0.92	0.75	1.75	0.14	1.94
Anxiety	0.05	0.33	0.16	0.71	0.00	0.00
Anger	0.08	0.46	0.22	1.23	0.00	0.00
Sadness	0.07	0.48	0.15	0.79	0.14	1.94
Family	0.09	0.56	0.20	1.03	0.00	0.00
Friend	0.36	1.43	0.55	1.52	0.11	1.54
Affiliation	1.55	2.80	2.61	4.15	0.88	6.80
Female	0.58	1.88	0.88	2.40	0.14	1.94
Male	0.16	0.76	0.53	1.74	0.14	1.89
Informal	0.25	0.94	2.07	4.35	0.38	3.37
Swear	0.00	0.00	0.09	0.57	0.07	1.34
Netspeak	0.20	0.88	1.57	3.92	0.09	1.20
Non-fluid	0.01	0.18	0.16	1.28	0.22	2.86
Filler	0.00	0.00	0.05	0.36	0.00	0.00

	HVI vs. AVI		HI vs. AVI		HVI vs. HI	
	p-value	Cohen's d	*p*-value	Cohen's d	*p*-value	Cohen's d
Word Count	0.03	0.16	0.00	1.45	0.00	1.50
Tone	0.04	0.17	0.00	1.37	0.00	1.14
Positive Emotion	0.78	0.05	0.00	0.69	0.00	0.67
Negative Emotion	0.00	0.37	0.61	0.07	0.00	0.33
Anxiety	0.00	0.22	0.30	0.20	0.00	0.35
Anger	0.03	0.16	0.27	0.24	0.00	0.27
Sadness	0.66	0.14	0.74	0.05	0.98	0.01
Family	0.08	0.14	0.13	0.23	0.00	0.29
Friend	0.25	0.13	0.06	0.17	0.00	0.29
Affiliation	0.02	0.31	0.15	0.13	0.00	0.30
Female	0.17	0.14	0.01	0.23	0.00	0.34
Male	0.01	0.29	0.98	0.01	0.00	0.21
Informal	0.00	0.62	0.84	0.05	0.00	0.44
Swear	0.33	0.25	0.48	0.08	0.94	0.02
Netspeak	0.00	0.52	0.77	0.11	0.00	0.54
Non-fluid	0.56	0.17	0.29	0.10	0.92	0.02
Filler	0.00	0.23	NA	NA	0.00	0.23

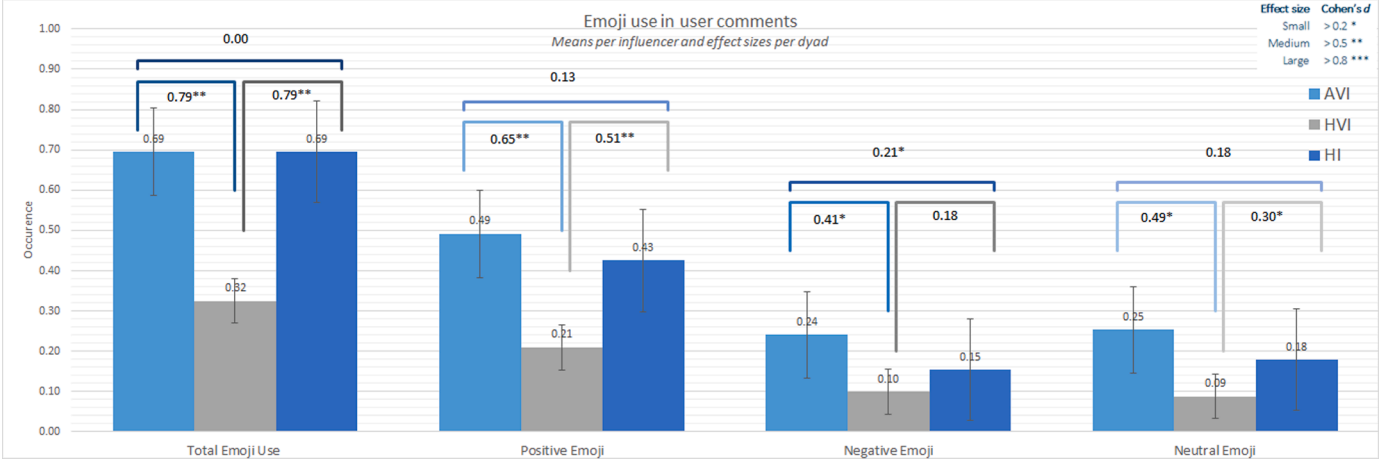


Fig. 2. Total, positive, negative, and neutral emoji use in user comments.

(Gilbert and Forney, 2015; Shechtman and Horowitz, 2003; Von der Pütten et al., 2010). However, our work goes a step further by also comparing the similarity of behaviour of virtual and human agents. Another novelty of our work lies in the real-life, publicly visible human network context, where a user community engages with a virtual agent, as opposed to much of the previous work on virtual agents where users are interacting with the agents principally in laboratory settings (Li, 2015). We introduced arguments both for and against users' willingness to engage with a virtual agent in a human network, and then investigated user reactions by analysing the behaviour of and reactions to two virtual influencers, one with a human-like and one with an anime-like representation, in comparison to an HI.

The virtual agents are anthropomorphized to varying degrees to facilitate interactions on social platforms, but this may in turn provoke perceptions of creepiness on the part of the human counterpart. We find that, of the two virtual agents in our study embedded in a publicly visible human network, the one with more human-like features elicited the least positive affect, with additional overtones of scepticism and

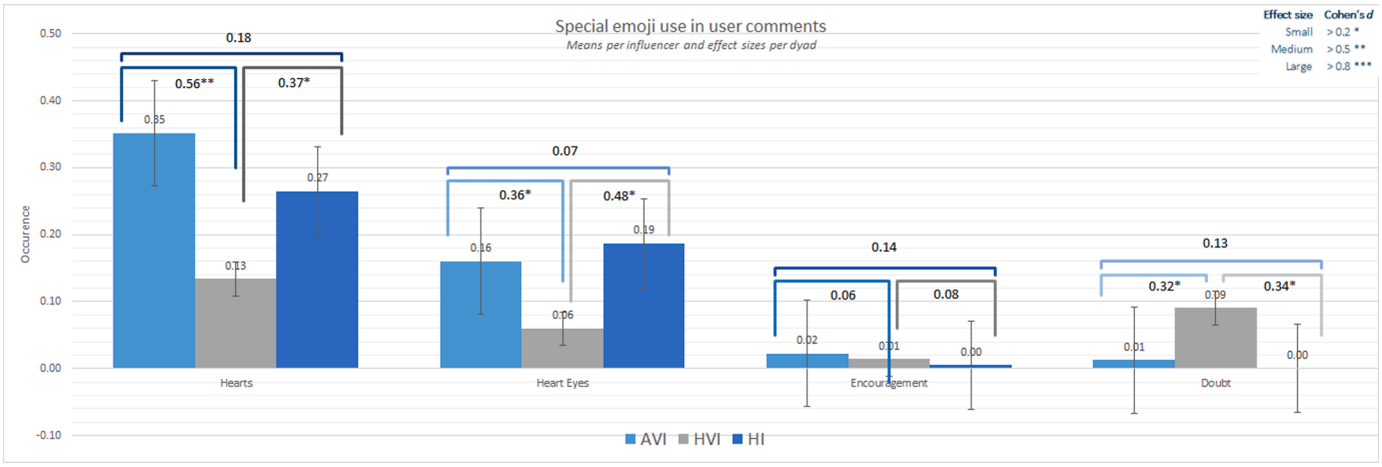


Fig. 3. Special emoji use in user comments.

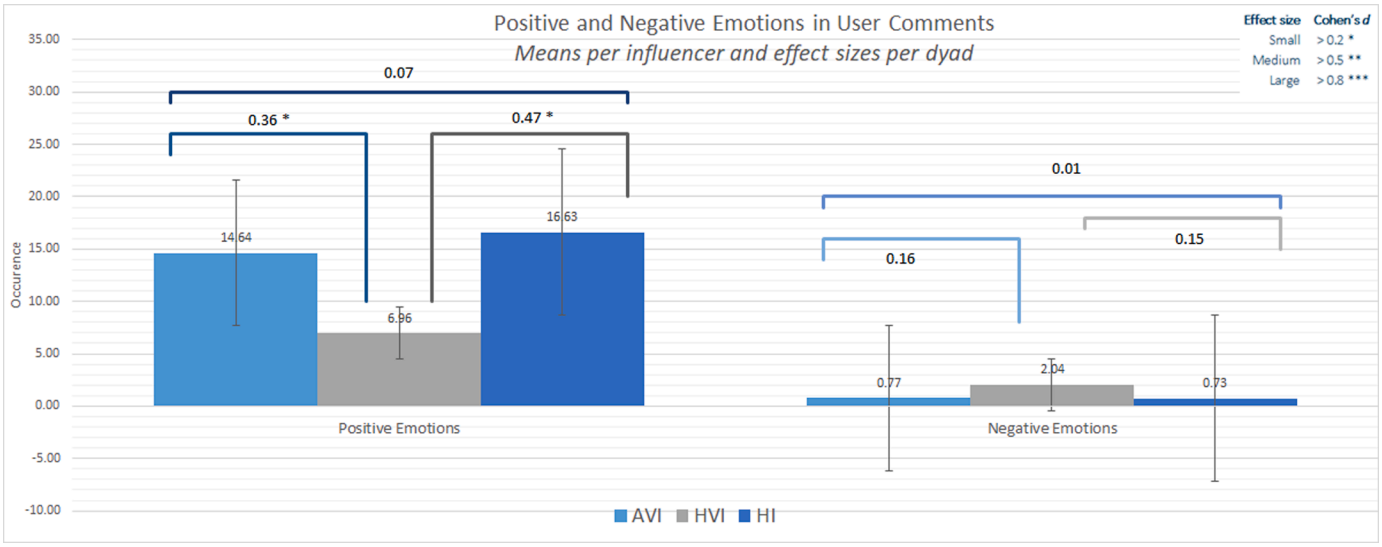


Fig. 4. Positive and negative emotions in user comments.

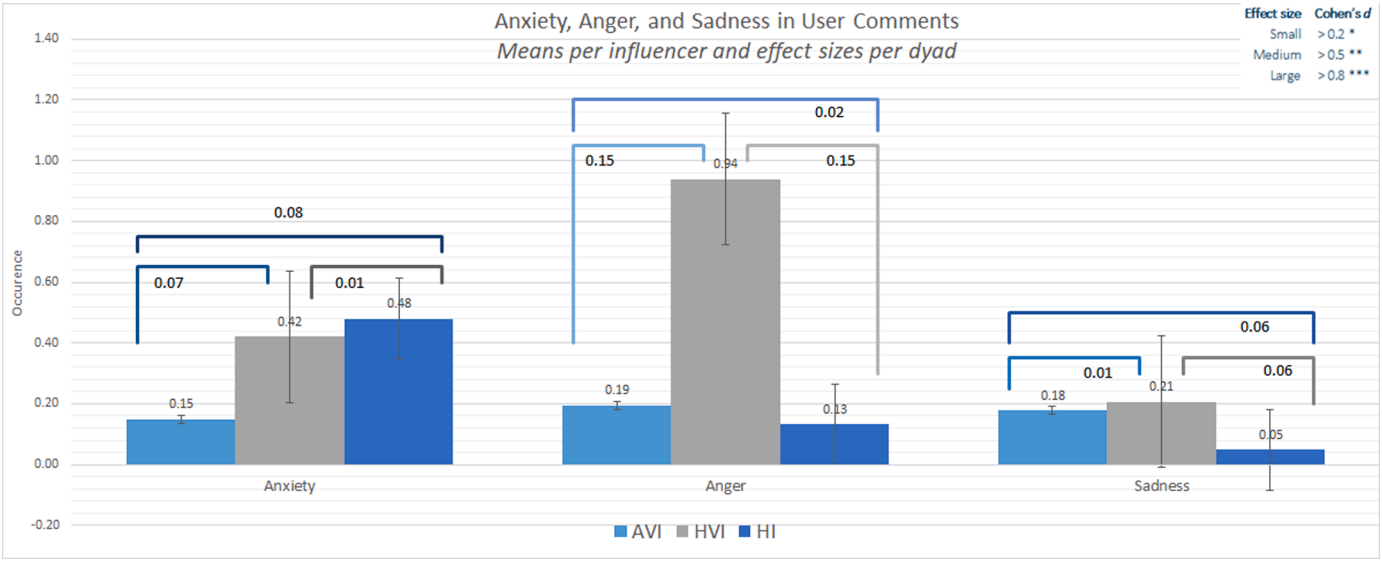


Fig. 5. Anxiety, anger, and sadness in user comments.

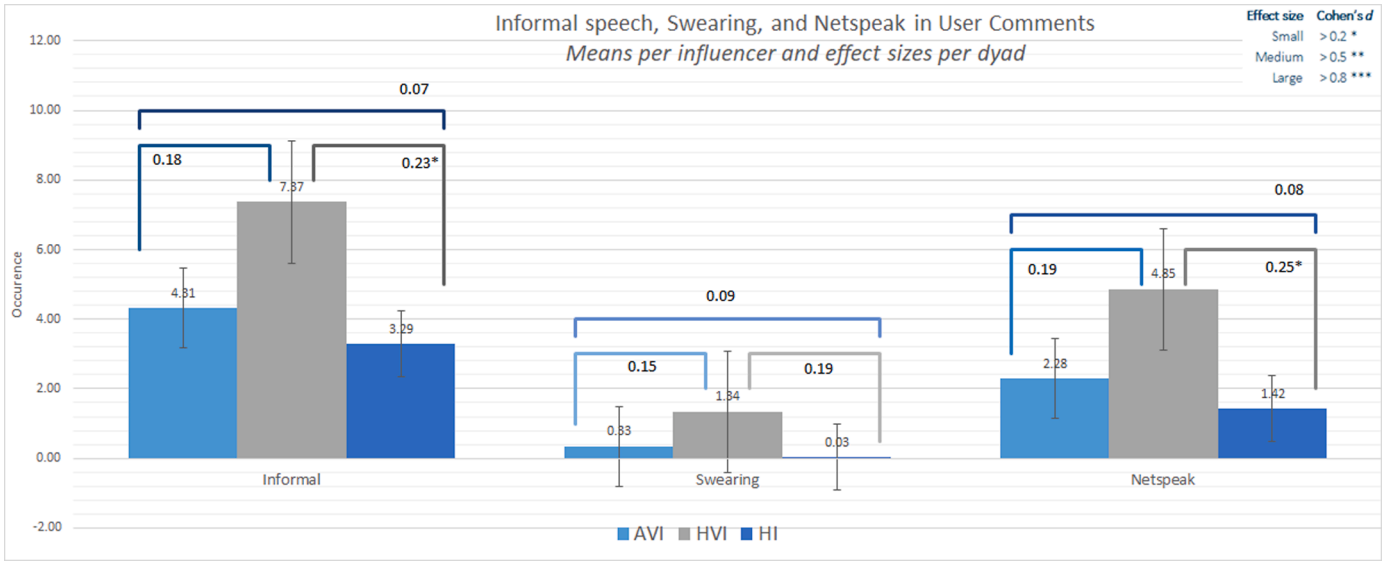


Fig. 6. Use of informal speech in user comments, with Swearing and Netspeak sub-categories.

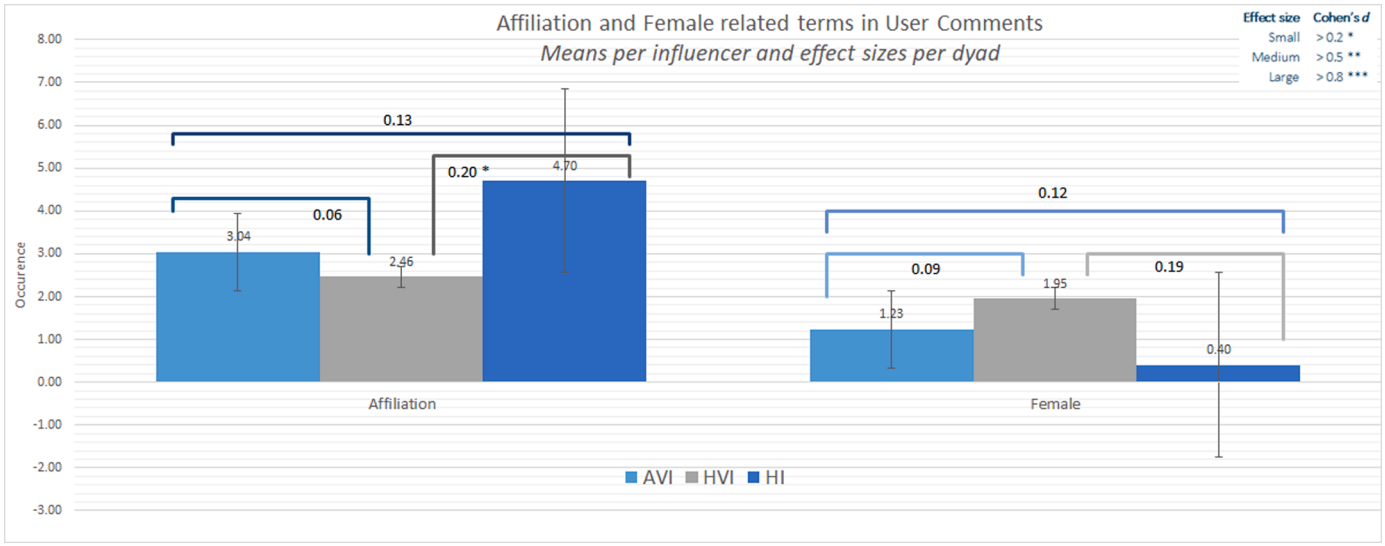


Fig. 7. Use of relation and gender terms in user comments.

dislike for her "robotic" nature, despite the other one being overtly CGI. In contrast, reactions to HI vs AVI showed very few differences. In general, it appears that the HVI's human-like nature means that, by appearing too similar to a human, or attempting to create a human-like experience for her followers, she elicits the expected experiences of creepiness, or negative emotional or cognitive reactance (Farman et al., 2020; Katsyri et al., 2017; Stein et al., 2020). We also identify the potential existence of an "Uncanny Hill" effect in user expressions of doubt, negative emotions, anger, and use of swear words.

Despite arguments for why virtual influencers may be treated similarly to human ones, we did not find substantial evidence for CASA tendencies in our data. However, the absence of significant differences between the three influencer types for some linguistic categories indicates that this is an area warranting future exploration.

5.1. *Uncanny valleys, uncanny hills*

We expected that virtual influencers would fulfil many of the same needs as HIs, including signalling one's social identity, diversion, and building personal relationships (Sheldon and Bryant, 2016). The virtual influencers aimed to create an atmosphere of pleasant diversion for their followers, with more positive overall tones, using more videos and emoji, and including more positive emotions in their posts. This may build intrinsic enjoyment of engagement with the virtual influencers, which in turn should satisfy needs of intrinsic motivation and therefore increase user engagement (Seaborn and Fels, 2015). Accordingly, the HVI received more likes, lengthier comments, and more video views, potentially creating an escapist fantasy for her audience (Partala, 2011), while the AVI received the greatest number of comments. The HVI received the most views of her videos, which may point to a curiosity effect of users wanting to see how an ambiguously human persona behaves. Interestingly, while the other two influencers avoided the use of negative content, the HVI does not show this reluctance, neither in terms of negative emoji usage nor expression of negative emotions.

Despite more engagement and more positive tone from the HVI, we see evidence of the UV when comparing the three influencers in terms of our indicators of positive affective reactions: positive emoji, heart emoji, heart-eyes emoji, and positive emotions. Interestingly, this is in spite of the general positive climate associated with Instagram content (Alhabash and Ma, 2017; Waterloo et al., 2018). We see this as strong support

Table 4
Emoji analysis of user comments.

# User Comments	AVI 12891		HVI 30536		HI 5402	
	M	SD	M	SD	M	SD
Emoji Use	0.69	0.46	0.32	0.47	0.69	0.46
Positive Emoji	0.49	0.50	0.21	0.41	0.43	0.49
Negative Emoji	0.24	0.43	0.10	0.30	0.15	0.36
Neutral Emoji	0.25	0.43	0.09	0.28	0.18	0.38
Hearts	0.35	0.48	0.13	0.34	0.27	0.44
Heart-eyes	0.16	0.37	0.06	0.24	0.19	0.39
Encouragement	0.02	0.15	0.01	0.12	0.00	0.07
Doubt	0.01	0.11	0.09	0.29	0.00	0.02

	HVI vs. AVI		HI vs. AVI		HVI vs. HI	
	p-value	Cohen's d	*p*-value	Cohen's d	*p*-value	Cohen's d
Emoji Use	0.00	0.79	1.00	0.00	0.00	0.79
Positive Emoji	0.00	0.65	0.00	0.13	0.00	0.51
Negative Emoji	0.00	0.41	0.00	0.21	0.00	0.18
Neutral Emoji	0.00	0.49	0.00	0.18	0.00	0.30
Hearts	0.00	0.56	0.00	0.18	0.00	0.37
Heart-eyes	0.00	0.36	0.00	0.07	0.00	0.48
Encouragement	0.00	0.06	0.00	0.14	0.00	0.08
Doubt	0.00	0.32	0.00	0.13	0.00	0.34

for the UV, as users clearly react differently, at least with respect to positive affective reactions, to the more human-like virtual agent than to either the AVI or HI.

Although the original UV hypothesis focused on emotional reactions on the positive affect spectrum, we also examined indicators of negative affective reactions to see if there was a corresponding "Uncanny Hill" – that is, greater negative affective reactions to the ambiguous influencer. In fact, we found evidence of this uncanny hill when looking at the negative emotions expressed in user comments, and specifically anger and swearing language. Given that positive and negative affect are not opposite ends of a spectrum but rather orthogonal constructs (Watson et al., 1988), the presence of a reverse effect is notable.

Although the pattern of negative reactions to the HVI may be a reaction to her own posting of more negative content, the pattern observed in the doubting category offers a potential refutation of this possibility. It is well acknowledged that many influencers post staged material (Djafarova and Rushworth, 2017), and the AVI is clearly not a real human. Yet it doesn't appear that either of these influencers merited high use of terms such as "fake" and/or "robot", the latter being particularly well suited to the concept of mechanistic dehumanization (Haslam, 2006). Contrary to research claiming that anthropomorphized embodiment of the virtual agent elicits sympathetic social behaviour (Luger and Sellen, 2016; Tan and Liew, 2020; Von der Pütten et al., 2010), this result implies that the interactive effects of virtual agent embodiment and claimed human-like mind, coupled with the virtual influencer's autonomous nature, may elicit negative reactions from users (Stein et al., 2020). This may be indicative of "agency tension", where a supposedly inanimate object (in this case a virtual agent) displays agency when interacting with the human. Interestingly this effect has been found to be offset by the presence of anthropomorphic cues (Kang and Kim, 2020), which appears to be the case for the AVI, yet not for the HVI.

It is also important to note that when looking at all of these positive

Table 5
Linguistic analysis of user comments.

# User Comments	AVI 12889		HVI 30536		HI 5402	
	M	SD	*M*	SD	*M*	SD
Word Count	3.04	5.46	5.59	7.92	1.69	2.74
Tone	38.42	41.56	37.04	33.58	36.01	42.06
Positive Emotions	14.64	28.04	6.96	18.04	16.63	31.08
Negative Emotions	0.77	5.51	2.04	8.79	0.73	7.27
Anxiety	0.15	2.29	0.42	4.08	0.48	6.45
Anger	0.19	2.60	0.94	5.84	0.13	2.79
Sadness	0.18	2.60	0.21	2.76	0.05	1.27
Family	0.28	3.67	0.41	4.08	0.11	2.78
Friend	0.58	4.97	0.64	5.96	0.77	7.76
Affiliation	3.04	11.62	2.46	9.60	4.70	16.67
Female	1.23	7.81	1.95	8.43	0.40	4.93
Male	0.08	1.45	0.42	3.43	0.06	1.79
Informal Language	4.31	16.03	7.37	17.89	3.29	14.82
Swearing	0.33	4.10	1.34	7.58	0.03	1.54
Netspeak	2.28	11.20	4.85	14.52	1.42	9.56
Nonfluid Speech	0.39	4.36	0.50	4.40	0.39	4.70
Filler Words	0.03	1.57	0.07	1.91	0.01	0.42

	HVI vs. AVI		HI vs. AVI		HVI vs. HI	
	p-value	Cohen's d	*p*-value	Cohen's d	*p*-value	Cohen's d
Word Count	0.00	0.35	0.00	0.28	0.00	0.53
Tone	0.00	0.04	0.00	0.06	0.14	0.03
Positive Emotions	0.00	0.36	0.00	0.07	0.00	0.47
Negative Emotions	0.00	0.16	0.94	0.01	0.00	0.15
Anxiety	0.00	0.07	0.00	0.08	0.58	0.01
Anger	0.00	0.15	0.71	0.02	0.00	0.15
Sadness	0.53	0.01	0.01	0.06	0.00	0.06
Family	0.01	0.03	0.02	0.05	0.00	0.08
Friend	0.54	0.01	0.13	0.03	0.34	0.02
Affiliation	0.00	0.06	0.00	0.13	0.00	0.20
Female	0.00	0.09	0.00	0.12	0.00	0.19
Male	0.00	0.12	0.95	0.01	0.00	0.11
Informal Language	0.00	0.18	0.00	0.07	0.00	0.23
Swearing	0.00	0.15	0.01	0.09	0.00	0.19
Netspeak	0.00	0.19	0.00	0.08	0.00	0.25
Nonfluid Speech	0.05	0.03	1.00	0.00	0.23	0.02
Filler Words	0.11	0.02	0.75	0.02	0.07	0.03

and negative categories, the difference between the AVI and the HI are negligible while the differences between each of these two influencers and the HVI influencer are much stronger. This provides further support for the UV hypothesis: it is not just a matter of people reacting more positively to HIs, but rather a clear indication that the human-*like* nature of the virtual influencer is eliciting a different affective reaction than a clearly non-human or clearly human representation. Apparently, appearing too realistically human does not provide an advantage (Katsyri et al., 2017; Wiese and Weis, 2020).

5.2. Faking authenticity

There is the obvious question of why the HVI would demonstrate such higher negative affect posting behaviour. Literature suggests that influencers need to pay close attention to the authenticity of their brand (Audrezet et al., 2018), and a profile that is always positive, without depicting any downsides of real life or personal flaws, may be seen as less relatable or interesting (Bareket-Bojmel et al., 2016; Djafarova and Rushworth, 2017; Johnson et al., 2019). Additionally, negative emotions are seen as more intimate, playing a role in the portrayal of one's true self-concept (Waterloo et al., 2018) and are a uniquely human experience, as computers neither react to nor retaliate against affective experiences (Wesche and Sonderegger, 2019). We argue that, as she walks the line between virtuality and humanness, displaying negative emotions adds to the HVI's attempts to "pass" as human, whereas the HI, with no concern about establishing her humanness, can more authentically display positivity in line with the general Instagram climate. Similarly, the AVI may not see a need to pass for a HI, and is therefore free to provide a more positive, escapist experience to her audience.

Another important characteristics of the anthropomorphization process and, by extension, attempts to pass for human, is the use of vocal displays (Edwards et al., 2019; Feine et al., 2019; Purington et al., 2017). Both virtual influencers rely more heavily on videos than the HI, which include them speaking or singing. They also demonstrate many of the social cues that incite people to interact with virtual agents (Tan and Liew, 2020) such as being female, stylish clothing, appropriate backgrounds, facial expression, attractiveness, emoji use, and photorealism (Feine et al., 2019). However, by attempting to infuse the HVI with human-like affective experiences, there is a risk of triggering excessive feelings of uncanniness if users perceive the virtual agent to be trespassing on what makes humans unique (Gray and Wegner, 2012; Stein et al., 2019).

Finally, the fact that the HVI uses more emoji in general than the other two influencers is not surprising. Previous research has found this to be a hallmark of the social media posting behaviour of both women (the majority of users on Instagram; Sheldon and Bryant, 2016) and influential social media users (Bogomilova, 2016; SwiftKey, 2015). Given the nature of the cases in this study and the platform considered, the higher emoji use by the HVI may represent an attempt to signal her identity or desired status.

5.3. Scepticism and antagonism

Not surprisingly, comments in response to the HVI showed more doubt about her true nature, questioning if she is "real" or "a robot". Again, the pattern of results across the three influencers is instructive: there were virtually no such sentiment toward the HI and while the AVI elicited some doubt, it was minimal and significantly below that expressed toward the HVI.

Although such doubts are understandable, this behaviour is curious as, by the time of this data collection, the HVI was well known for being non-human. Questioning her virtual nature is at best frivolous, at worst purposefully derogatory, especially when this information is readily available in the HVI's Instagram bio. Therefore, the act of posting such doubts may be evidence of attempts at mechanistic dehumanization of the HVI (Haslam, 2006) and may stem from a reverse social identity effect, where users want to disassociate themselves from her audience. When evaluating an entity considered to belong to a different group than the evaluator, group characteristics with which the evaluator identifies highly become more salient (Edwards 2019). In our case, the HVI displays stereotypical traits that should incorporate her into the social groups of many social media users (e.g., young, female, stylish) and yet she still elicits a need for disassociation, as opposed to the overtly non-human influencer. Such disassociation may play a protective role, explicitly distancing the user from negative stereotypes associated with membership in a given group (Tajfel, 1978): despite her similarities to other users, there may be a desire to clearly delineate the human-like agent as "not human". This may reflect an individual difference that makes some users particularly sensitive to stimuli that elicit eeriness, a hypothesis that we were not able to test with the current data set.

Our analysis showed that users responding to the HVI used more female-related terminology compared to the HI. Although the HVI is non-human, she is clearly represented as a woman, therefore there should be no need to question her gender; the AVI (also clearly presenting as a female) also received more female related terminology than the HI. We believe that this reflects another aspect of the doubting phenomenon, where users are questioning whether the virtual influencers are "real" women. The believability of a virtual agent relies very much on gender stereotypes, and virtual agents are more likely to be presented as overly feminine (Brahnam and Weaver, 2015). We contend that this gender identification may have triggered gender stereotypes surrounding expected behaviour (Brahnam and Weaver, 2015; Feine et al., 2019). It is possible that the seemingly cold and superficial nature associated with mechanistically dehumanized objects may violate female stereotypes of warmth and caring (Brahnam and Weaver, 2015; Haslam, 2006), making this a salient characteristic of focus for some users.

Online contexts are ones where social norms are not yet fully developed and antisocial behaviour, such as trolling and online harassment, is frequently observed (Cheng et al., 2017). On the one hand, Shechtman and Horowitz (2003) provided evidence that users are more prone to hostility towards human partners than to computers in assertive conditions. On the other hand, the dehumanization of a non-human agent may make it easier to engage in unpleasant interactions with them (Haslam and Loughnan, 2016). In either case, prior research has found that negative behaviours in virtual environments can be contagious (Cheng et al., 2017). Therefore, we had reason to expect to see equally antisocial reactions - in the form of negative emotion, anger, and swearing - to all three influencers. However, this was not the case. There was a clear trend of increased negative and antagonistic reactions to the HVI, with again little difference between the AVI and HI. Even though at least part of this pattern may be attributed to the influencers' own posting behaviours, it is still curious that, given the above arguments for why antagonistic behaviour should be more prevalent online, there is clearly a difference between the HVI and the other two. Viewed from a different perspective, it appears that the positive norms of Instagram did not have the same buffering effect for all three influencers.

Another possible explanation for these results is that users were purposely testing the limits of the HVI. Either due to the ambiguity surrounding the driving force behind the virtual influencer profiles or their true nature, users may have attempted to engage in shocking or norm-breaching behaviour (e.g. using more swear words) to gauge how the virtual influencer would react. If this is the case, however, it is curious that this antagonistic behaviour was not the same for the two virtual influencers, implying that any such testing phenomena was a product of the HVI's strong, yet still imperfect, resemblance to a human rather than questions about the technology (or humans) behind the posting behaviour.

5.4. Almost human

Social connectedness is a fundamental human need, one that is being increasingly met via digital technologies (Gaines, 2019; Partala, 2011) and research has shown that physical presence is not necessary for the experiencing of social presence (Von der Pütten et al., 2010). In terms of physical manifestations of the HVI's humaneness, her creators have already anthropomorphized her as much as possible in terms of physical appearance – in contrast to the AVI that clearly presents anime features. Additionally, the storylines and multimedia content developed for the HVI's profile attempt to mimic human experiences of friendship, with both real and virtual others – again, as opposed to the AVI, who poses with other humans primarily for professional purposes, such as modelling. We therefore look more closely at some linguistic categories that allow us to consider whether the virtual influencers garner the same reactions as the HI.

To look for evidence of CASA effects, we looked at terminology that would imply a social connection being established between the user and the influencer. We saw no difference in the use of terminology associated with affiliation between the two virtual influencers, and no difference in family and friendship related terminology across the three influencers. The lack of a clear pattern in these results leads us to conclude that we cannot reject the presence of CASA effects in our data.

One potential consequence of a platform such as Instagram, which promotes filtered, staged, and perfected lifestyles, may be social comparisons and resulting feelings of anxiety for not being able to achieve the same level of perfection as that displayed by the HIs (Djafarova and Rushworth, 2017). In our results, anxiety did not display neither of the patterns observed in other variables. Although the differences between the three influencers were not large enough to cross even the small effect size threshold, comments in response to the HI saw the highest score, while those in response to the AVI were the lowest. The fact that a virtual influencer appears to be living an idealistic lifestyle may actually cause lower anxiety in users, as such a lifestyle appears unachievable for reasons other than their own personal shortcomings. This effect was particularly strong for the AVI who, although displaying flawless skin, makeup, and fashion choices, is clearly not real and therefore may be an obviously irrelevant comparison target.

6. Conclusion

We provide several insights for embedding virtual agents in human networks, particularly in online communities consisting of multilateral communication. Firstly, specific to the context of this study, we propose that virtual agents may serve as a viable substitute for influencers, overcoming the ethical and practical issues discussed earlier (Khamis et al., 2017). Even prior to the COVID-19 pandemic, increasing negative perceptions of influencers were fuelling concerns about the sustainability of the influencer lifestyle. In the wake of the current pandemic-induced economic crisis facing many brands and consumers (Cellan-Jones, 2020), virtual influencers may present a unique opportunity (Tan and Liew, 2020). For one, as they are not affected by anxiety, isolation, or depression caused by uncertainty, they may be able to continue creating regular content. Additionally, although social media has come to mitigate the lack of viability of other advertising forms in the current climate, the glamor and cheerfulness traditionally associated with influencer posts is now less acceptable (Suciu, 2020); however a virtual influencer's "authentic" fakeness may allow them to post content that serves as a distraction for users. For example, LilMiquela "visited" a deli during the pandemic for an advertised performance[16]. Her virtual nature allows her to not only bypass any physical or legal restrictions, but potentially also to exploit the clearly fictional nature of her existence. Alternatively, virtual influencers can offer a sustainable source of influencer marketing that can be easily "turned off" without concerns for the wellbeing of the person behind the profile. Attention should still be paid to the level of the authentically "human" experience required by a given brand to determine whether user reactions can easily translate from a human to a virtual influencer.

When introducing virtual agents into human networks, careful attention must be paid to the balance between providing the users with an experience similar to that offered by human agents, while not making the virtual agent too autonomous or human-like. One of the virtual influencers in our study is purposely designed to be extremely human-like, not only in her appearance but also her posting behaviour and the content of her photos and videos, even leaving much of the online world guessing as to her real nature for some time. And yet, she cannot avoid negative reactions tied to being "not real", as opposed to her anime-like counterpart. Practitioners may be better advised to maintain some element of "virtualness" in their virtual agents, rather than attempting to maximize their humaneness, recognizing that this objective may be at odds with the limits of human psychological reactions to non-human entities.

6.1. Limitations and perspectives

Some limitations must be considered given the nature of our study and data collected. Firstly, Instagram's API restrictions noticeably limited our access to the complete pool of comments per post. We were nevertheless able to collect a large sample for each influencer and have no reason to believe there is any systematic bias.

Furthermore, given the limited number of virtual influencers available and the variability of their nature, this paper considered only three cases of influencers. These results may not be generalizable to virtual agents targeting different demographics, operating in different content categories, or across different digital medium. For instance, reactions may differ on a more textual medium such as Twitter (Sheldon and Bryant, 2016), as opposed to the highly visual medium of Instagram (Djafarova and Rushworth, 2017; Johnson et al., 2019). Research should be expanded to consider different context and individual characteristics, and to develop potential contextual boundary conditions. However, the current limited availability of such agents restricts the potential foci and sample size of such studies. Moreover, image semiotics may be used to further analyse the visual stimuli on Instagram to interpret the societal or cultural contexts within which such reactions and interactions are occurring, in addition to the textual interpretation.

Although users are signalled that these personas are not real, the exact mechanics of the virtual influencers' behaviour and their content creation is unclear. The current study was unable to examine whether users believe they are interacting with a human being behind an avatar or an AI entity. Future research may explore the effects of users' understanding of the virtual agent's true nature.

Analysing virtual agents engaging in human activities (performing arts, political commentary, etc.) offers another research premise to gain insight into user reaction when presented with varying levels of human-like content from a non-human entity. Such studies could explain how virtual agents can be effective opinion leaders - outside of the lifestyle realm - on crucial topics such as health, career advice, or social activism. These results could inform companies, NGOs, and government agencies as to how to leverage virtual agents as communicators and ambassadors.

Research into emoji use and related sentiments is still new, and studies show that emoji can have different affective valences based on the social and cultural context, attached text, operating system used, author identity, etc. (Fernández-Gavilanes et al., 2018; Tigwell and Flatla, 2016). Additionally, emojis are more frequently used to convey positive emotions than negative ones and users tend to prefer anthropomorphic symbols (Bogomilova, 2016; SwiftKey, 2015). One limitation of our analysis is that some emoji were present across all three categories (positive-neutral-negative), which may explain the similar pattern of emoji use. However, we added heart and heart-eyes emojis to provide an

[16] https://www.instagram.com/p/CGKy7GZDgou/?igshid=17vpox4uj2r46

independent evaluation of positive reactions, as these were removed from the positive emoji list prior to analysis. Therefore, our results showing potential uncanny hill effects of negative reactions warrant further study. Negative reactions received by the HVI may be explored to understand if they are a response to her greater use of negative emotions, or a more targeted reaction to either her uncanny/ambiguous nature (Wiese and Weis, 2020), or her attempts at faking authenticity.

Given the results showing ambiguous CASA results, further examination is required to understand whether the followers of virtual influencers build the same sorts of social relationships with them as with human ones. Future research may unearth similarities in parasocial relationship formation or social identity effects via tagging behaviour or the use of hashtags, which signal speaking directly to an influencer or identifying with similar interests. The current study was only able to investigate reactions of users who were already aware of the virtual influencers in some way – either following them or encountering their online activity. It would be interesting for future studies to consider non-followers' reactions, and how decisions regarding following or interacting may differ between virtual and human influencers.

The reverse social identity effects where we suspect that some users were actively attempting to disassociate the HVI from categories such as "human" or "female" should be further examined. Women, as the predominant gender on Instagram, may have the strongest incentive to question the "femaleness" of the virtual influencers in order to disassociate them from the "human female" in-group. Additionally, further exploration of feelings of affiliation and acceptance of virtual influencers into one's social identity is required, potentially building on the budding literature on parasocial relationships in social media contexts (Hwang and Zhang, 2018; Kim and Song, 2016; Sokolova and Kefi, 2020). Another promising area of research would be to investigate potential boundary conditions that may exist at which point highly anthropomorphised agents engender such reverse social identity effect, or put differently, at which point the descent into the UV begins (Mathur et al., 2020), particularly in contexts where interactions are visible to others.

In conclusion, it appears that a very human-like virtual influencer pays a price in terms of user reactions, which is not incurred by its anime-like, and clearly non-human, counterpart. The uncanny appearance, combined with attempts to mimic human posting and vocal behaviors, may precipitate the unavoidable descent into the UV as the lines between virtual and human blur. In order to avoid this, virtual agents in human networks may have to resign themselves to remaining almost – but not quite – human.

CRediT authorship contribution statement

Jbid Arsenyan: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Agata Mirowska:** Conceptualization, Methodology, Validation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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